

Thoracic Trauma

Guideline Title: Thoracic Trauma	Guideline Number: GUID-3203-T015
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CXXXVII. **PURPOSE:** To recognize the patient who has suffered thoracic trauma and treat life-threatening injury immediately. To provide the patient with adequate airway, perfusion, and pain management and transport to a facility with the ability to care for the patient.

CXXXVIII. **PATHOPHYSIOLOGY:** Thoracic trauma is classified as blunt or penetrating. Blunt trauma to the thorax is associated with vehicular collisions, falls, and assaults, whereas penetrating trauma occurs as a result of gunshot wounds, stab wounds, and impalement. Life threatening thoracic injuries may include: airway obstruction, tension pneumothorax, massive hemo-thorax, open pneumothorax, flail chest, cardiac tamponade, aortic rupture/transection, myocardial rupture, myocardial contusion, pulmonary contusion, tracheobronchial disruption, diaphragmatic disruption. A decrease in the blood volume, failure to ventilate the lungs, ventilation perfusion mismatches or pressure changes within the intrapleural space are contributing causes of hypoxia.

CXXXIX. **DEFINITIONS:**

CXL. **DEPARTMENT GUIDELINE:**

- A. Conduct primary assessment; major problems identified should be immediately treated.
- B. Assess for and control any external hemorrhage. Stabilize any penetrating objects.
- C. Assess and manage airway, breathing, and circulation.
 - i. Determine if patent or if there is a partial/full obstruction. Evaluate for adventitious airway sounds (i.e. stridor), increased secretions, blood in the airway.
 - ii. Assess respiratory quality, rate, spontaneous effort, presence of absence of breath sounds, chest excursion and expansion, c/o of dyspnea, position of trachea, external signs of chest trauma, bony or air crepitus upon palpation, paradoxical movement, presence of flail segment and/or presence of JVD.
 - iii. Assess SpO2 reading as well as ETCO2 if applicable.
- D. Assess for mechanism of injury, vehicle damage such damage to the steering wheel or steering column, +/- use of seatbelt, intrusion into the driver or passenger compartment, ejection of patients, height of fall and/or speed at which the pedestrian was struck.
- E. Assess for signs of shock, altered mentation, hypotension, or increased HR, and/or signs of peripheral vascular constriction. Monitor continuous vital signs.

- F. If hemodynamic instability, consider blood product transfusion per *GUID-PR025 - Emergency Blood Product Administration* or *GUID-PR002 Blood Product Administration* with *GUID-PR026 – Warming of Blood Products & Intravenous Fluids*.
- G. If signs of a tension pneumothorax (increased heart or respiratory rate, decreased SpO₂, decreased lung sounds to the affected side, JVD, subcutaneous emphysema, hypotension, or tracheal deviation), perform a needle thoracostomy per *GUID-PR017- Needle Thoracostomy*.
- H. If clinical signs of an open pneumothorax, place a sterile occlusive dressing, large enough to overlap the wound edges, that is taped securely on 3 sides. If signs of a tension pneumothorax develop, release occlusive dressing before performing a needle thoracostomy.
- I. Stabilize any noted flail segments. Consider positive pressure ventilation
- J. Provide adequate analgesics per *GUID-G005 – Analgesia and Sedation Management*.
- K. If interfacility transport
 - i. Assess initial CHEST RADIOGRAPH for signs of hemothorax/pneumothorax.
 - ii. If chest tube is in place, maintain closed chest tube drainage system. When feasible, maintain suction per *GUID-PR005 – Chest Tube Management*.
 - iii. If *confirmed* blunt aortic injury, consider maintaining HR<100 and SBP approx. 100mmHg with **Esmolol: IV bolus of 500 mcg/kg** over 1 minute then begin infusion at 50 mcg/kg/min. Titrate every 5 minutes by 50 mcg/kg/min. Max dose 300 mcg/kg/min.
 - 1. If esmolol unavailable, **Labetalol 10-20 mg slow IVP** every 10 minutes - Max total dose of 300 mg
 - iv. Obtain all pertinent paperwork, X-rays, and CT scans.
 - v. Consider obtaining a 12-lead EKG.
- L. In the setting of traumatic cardiac arrest with suspected chest trauma, consider *bilateral pleural decompressions* as part of the resuscitation efforts.

CXLI. **DOCUMENTATION:**

- A. EMS Charts

CXLII. **REFERENCES:**

- A. Fabian, T.C., Davis, K.A., Gavant, M.L., et al. (1998). Prospective study of blunt aortic injury: helical CT is diagnostic and antihypertensive therapy reduces rupture. *Ann Surg*, 277, 666.